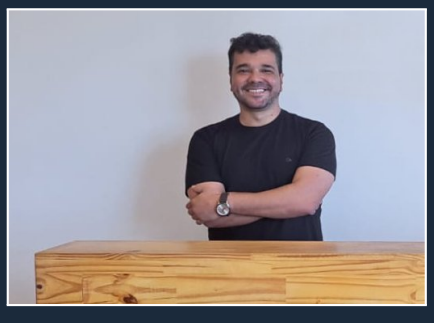
LUCAS ALVES DE SOUZA

Data Scientist

lucas-data-science.github.io/ lucas-alves-souza/ lucasfusj@gmail.com

Data Scientist who transforms complex data into actionable insights that support business decisions. I develop statistical models, machine learning pipelines, and strategic analyses, combining hands-on data experience with a strong quantitative background and a PhD in Physics.



Data Science Experience

Data Scientist – Specialist level — Tenda Construtora
9/2021 – present

- Data Scientist (specialist level) in Market Intelligence for residential real estate, supporting pricing, sales performance, sociodemographic analysis, and executive decision-making.
- Develops and validates machine learning and statistical models for dynamic pricing, demand analysis, price elasticity, sales velocity, and revenue potential.
- Builds dashboards and clustering models using sociodemographic, geographic, and market data to monitor performance, classify micro-regions, and identify actionable market opportunities.
- Translates complex analyses into clear recommendations and executive presentations for directors and senior stakeholders.

Education

2017 **Doctorate in Physics** — ITA
2011 **Master's degree in Physics** — UFSJ
2008 **Physics Undergraduate** — UFSJ

Selected Training (notes/scripts → Github)

2025 Statistics for Data Science and Machine Learning — IA Expert Academy / Udemy
2022 TensorFlow: Machine Learning and Deep Learning with Python — IA Expert Academy
2022 Genetic Algorithms in Python — IA Expert Academy
2021 SQL Bootcamp — Udemy
2021 Machine Learning and Data Science with Python — IA Expert Academy / Udemy
2017 School on Few-Body Physics — MPIPKS, Dresden, Germany

Study Notes (see Github page)

- Data Science → https://lucas-data-science.github.io/ml_notes.
- Generative AI for Data Science → https://lucas-data-science.github.io/GenAI_notes.

Recent Publications (see more)

- Lucas A. Souza, *Neural-network solution of subtracted three-body Faddeev integral equations near the Efimov limit,* Few-Body Systems, 67, 49 (2026).
- Lucas A. Souza, *Performance-Driven Causal Signal Engineering for Financial Markets under Non-Stationarity,* Trans. Quant. Fin. Bey. Vol. **3**, No. 2, 79-99 (2026) .

Research Experience

Postdoctoral Researcher — ITA
2020 Investigation on short-range correlations in nuclear matter of neutron stars.

Postdoctoral Researcher — USP
2019 Theoretical study on nuclear reactions of neutron-rich nuclei.

Technical Skills

Machine Learning & Statistics: Machine Learning, Statistical Modeling, Predictive Modeling, Feature Engineering, Model Validation, Model Evaluation, Hyperparameter Optimization, Quantile Regression. **Programming & Platforms:** Python, SQL, Databricks, Pandas, Big Data Processing, Data Pipelines, ETL/ELT. **Forecasting & Business Analytics:** Demand Forecasting, Time-Series Forecasting, Probabilistic Forecasting, Pricing Analytics, Backtesting, Advanced Analytics, Exploratory Data Analysis and Data Visualization. **Professional Competencies:** End-to-End Data Science, Scientific Communication and Data-Driven Decision Making. **Generative AI:** Large Language Models, Retrieval-Augmented Generation, Embeddings, Transformers, Natural Language Processing and Agentic AI.